ELEC70067
EE9-SO7

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE PART IV: MEng and ACGI

TRAFFIC THEORY AND QUEUING SYSTEMS

Monday, 28 April 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : D. Mandic

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Special information for students

1. Erlang Loss formula recursive evaluation:

$$E_N(\rho) = \frac{\rho E_{N-1}(\rho)}{N + \rho E_{N-1}(\rho)}$$
$$E_0(\rho) = 1$$

2. Engset Loss formula recursive evaluation (for a fixed M and $p = \alpha/1+\alpha$):

$$e_N = \frac{(M - N + 1)\alpha e_{N-1}}{N + (M - N + 1)\alpha e_{N-1}}$$
$$e_0 = 1$$
$$\alpha = \lambda/\mu$$

3. Kleinrock's conservation law:

$$\sum_{k=1}^{m} \rho_k E[W_k] = \left(\frac{\rho}{1-\rho}\right) E[R]$$

4. Expected residual time and waiting time:

$$E[R] = \frac{1}{2}\lambda E[S^2] = \frac{1}{2}\sum_{k=1}^{m} \lambda_k E[S_k^2]$$

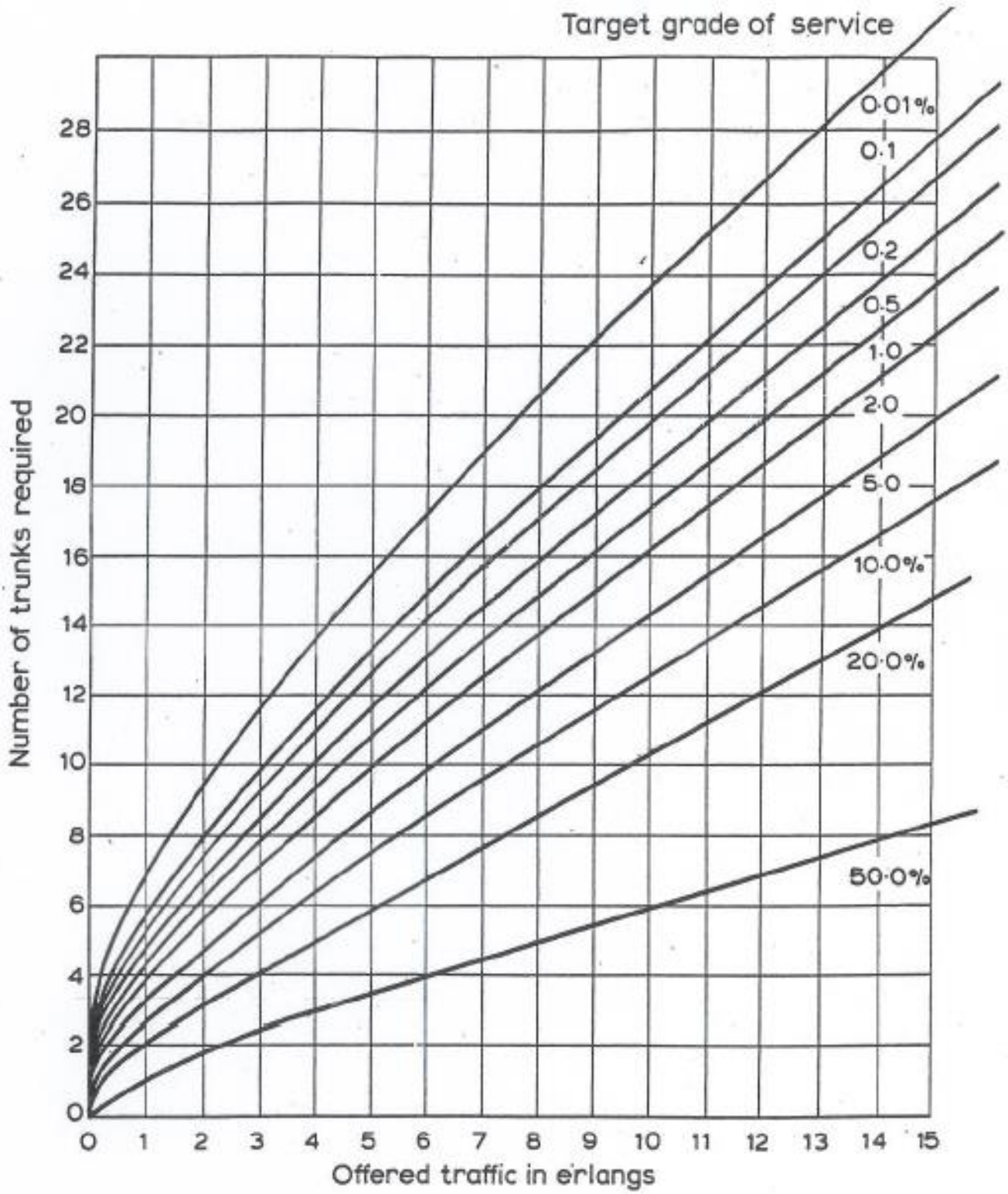
$$E[W] = \frac{E[R]}{(1-\rho)}$$

5. Mean and variance of the Kosten overflow model (infinite overflow link $N = \infty$):

$$m = E[N_t] = \rho E_M(\rho)$$

$$v = Var(N_t) = m\left(1 - m + \frac{\rho}{M+m+1-\rho}\right)$$

6. Traffic capacity on basis of Erlang B formula (next page).



Traffic capacity on basis of Erlang B. formula.

The Questions

1. Birth/death process and Erlang Chart.

a)

i) For an Erlang model prove that, if the Markov chain is in state, i , the rate of departure is $\mu_i = i\mu$.

Clearly highlight all the steps in your derivation. [4]

ii) Discuss the relationship between the Erlang and Engset models, using the expression $\lambda_i = (M - i)\lambda$ for the birth rate, when in state i of an M-sources Engset model. [3]

iii) Requests to a call centre arrive at a mean rate of 1 call per second. The mean call holding time is 15 seconds. The centre is designed to have a loss probability $B_c \approx 1.0\%$.

How many channels will the centre need to be equipped with to comply with the loss probability specification? [3]

b) A 6-channel communication link is offered a mixture of two types of traffic.

Traffic A with parameters (λ_1, μ_1) requires one (1) channel to transmit a message. Traffic B with parameters (λ_2, μ_2) requires simultaneously three (3) channels to transmit a message.

i) Derive the $2 - D$ birth/death model for the traffic on the link. [5]

ii) Prove that the equilibrium traffic distribution has a product form. [5]

Hint: verify that the local balance equation holds for this system.

2. IPP model and overflow.

a) Pure chance traffic is offered to a 15-channel communication link. The traffic parameters are $\lambda = 3$ calls/minute and $\mu = 1/3$ calls/minute.

Blocked calls are accepted into a 1-channel overflow link.

i) Using Kosten's overflow model derive the mean and the variance of the overflow traffic. [4]

ii) You are asked to derive an equivalent IPP overflow traffic model shown in **Fig. 2.1.** Derive all the parameters of the model. [4]

Hint: you can use the following results for an IPP Markov chain model ($N = \infty$):

$$E(N_t) = \left[\frac{\alpha}{(\alpha+\beta)}\right]\rho$$

$$E\big(N_t(N_t-1)\big) = \left[\frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}\right]\rho^2$$

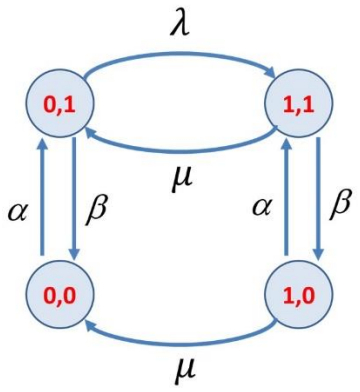


Figure 2.1

2. M/M/K/N system

b) A Poisson stream with a mean rate of arrival of 7200 calls/minute is offered to a system that allows arrival to wait in a buffer. The size of the buffer B of the system is three (3) and the capacity of the system ($N = K + B$) is five (5).

Each server can serve calls at a rate of 100 calls/second.

For this system the set of local balance equations is given by:

$$\pi_i = \left(\frac{A^i}{i!}\right)\pi_0 \text{ for } 0 \le i \le K$$

$$= \left(\frac{A^K}{K!}\right)\rho^{i-K}\pi_0 \text{ for } K \le i \le K + B$$

$$\pi_0 = \frac{1}{(A^K/K!)}\left[\frac{(1-\rho)E_K(A)}{(1-\rho)+\rho(1-\rho^B)E_K(A)}\right]$$

i) Show that: $P[delay] = \pi_K\left(\frac{1-\rho^B}{1-\rho}\right)$. [5]

ii) Show that: $P[loss] = \pi_K\rho^B$. [3]

For this system:

iii) Derive the probability that an incoming call is lost. [2]

iv) Derive the probability that an incoming call is delayed. [2]

3. Non-Markov queuing system.

a) The arrival pattern to a single point of service (server) is governed by a Poisson distribution. The arriving requests are a random mixture of jobs of fixed length:

1/3 of the jobs are of length B bits, 1/4 of the jobs are of length 2B bits, and 5/12 of the jobs are of length 3B bits.

Assume that the job size B = 160 bits, the mean rate of arrival is $\lambda = 150$ messages/second, and that the speed of the server is $\mu = 64$ kilobits/second.

i) Determine the mean service time experienced by a call. [2]

i) Determine the mean waiting time experienced by a call. [5]

ii) Determine the mean transmission time of a call. [2]

b) Equivalent capacity approximation.

i) Define and describe the steps to derive an equivalent capacity when the number of sources is large, and the effect of the access buffer is neglected. [4]

ii) Define and describe the steps to derive an equivalent capacity that considers the effect of the access buffer. [4]

iii) Discuss in your own words potential applications of the information provided by an equivalent capacity function. [3]

4. Fluid flow model and Performance/Reliability analysis.

a) For the fluid flow model with N sources, shown in **Fig. 4.1.**, a well-known approximation to the probability that the buffer occupancy exceeds x is given by:

$$G(x) = A_N \rho^N e^{-Rx}$$

where R is the dominant smallest negative eigenvalue.

i) Discuss the key steps of the derivation of the statistical equilibrium equation for the fluid flow model in **Fig. 4.1.** [5]

ii) Using a suitable (and small) value of N, show that the solution to this system takes the form of a sum of exponentials. [5]

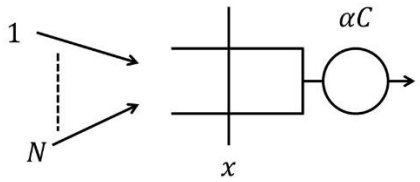


Figure 4.1

b) A two-component system is operational if at least one component is working.

The system is not operational if both components are in fault.

i) Define and identify the four states of this system. [2]

ii) Individual components can reboot to their operation state - but only if the system is still operational. Identify the transitions that represent this policy in the state space diagram. [2]

Note: In this situation maintenance personnel will mobilise immediately to replace the faulty system on site.

iii) Using the diagram derived in ii), include the transition(s) that will represent a reboot to a (fully operational) stand-by system. [3]

Note: In this case, maintenance personnel are periodically visiting sites and replacing faulty standby systems.

iv) Briefly discuss how would you decide between the policies ii) and iii) in terms of system availability and associated operational costs. [3]